

From Calibration to Closure in a Compton-Core Finite-Filament Model: Spherical and NLS Pressure-Cell Closure for the Fine-Structure Scale

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Abstract

A previous Compton-core finite-filament scaling model showed that several electron-scale quantities can be fixed algebraically once a single dimensionless core aspect scale is supplied. This note develops the intermediate pressure-cell closure needed to replace that external aspect scale by a finite-cell eigenvalue in the companion BEM/NLS calculation. The local Biot–Savart logarithmic coefficient is reviewed in the slender-core normalization, leading to the no-contact core condition $a_{\text{nc}} = r_c$ when $A_K \rightarrow 1/(4\pi)$. A spherical control-volume response gives

$$\Xi_{\text{sph}} = 1 + \frac{3}{4\mathcal{L}_K} + \frac{1}{16\mathcal{L}_K^2}.$$

The main new ingredient is the NLS finite-shell pressure scale

$$E_p^{\text{NLS}} = \frac{16\pi}{3}\mathcal{L}_K \left(1 - \frac{11}{48\mathcal{L}_K^2}\right),$$

together with the shell stiffness

$$\kappa_{\text{NLS}}^{\text{shell}} = 8\pi\mathcal{L}_K^2.$$

For the ideal trefoil value $\mathcal{L}_K = 16.371637$, these give $E_p^{\text{NLS}} = 274.0748756568$ and $\kappa_{\text{NLS}}^{\text{shell}} = 6736.341149$. These quantities are not obtained from α , m_e , $\hbar$, e , or R_∞ ; they are geometric/NLS inputs for the companion finite-cell BEM eigenvalue problem. The present paper therefore supplies the analytical pressure-cell and NLS-shell closure. The numerical extraction of the stationary aspect eigenvalue $E_\star$, and the far-field identification with the electromagnetic coupling, are treated as companion results.

1 Purpose and status

The objective is to separate a numerical calibration from a mathematical closure. Let $\mathcal{E}_0$ denote a zeroth-order topological aspect scale and let $\mathcal{L}_K = L_K/D$ denote the ropelength of a diameter- D ideal trefoil centreline. The present paper proves the following conditional result:

Given the finite-core Biot–Savart self-energy, the circulation relation $\Gamma_0 = 2\pi r_c \|\mathbf{v}_\odot\|$, the spherical pressure-cell functional of Sec. 6, and the aspect-renormalization rule of Sec. 8, the numerical fine-structure scale is fixed by $\mathcal{E}_0$ and $\mathcal{L}_K$.

The proof is conditional because $\mathcal{E}_0$ itself is not derived here from primitive field equations. Nor is the far-field electromagnetic $1/r$ law derived. Those are promoted to explicit open theorems in Sec. 10. This keeps the paper falsifiable: the model succeeds as a closure theorem if the stated assumptions are accepted, and it becomes a fundamental derivation only if those assumptions are later obtained from a unique dynamical principle.

2 Derived-label gate and assumption ledger

The word “derived” is reserved here for quantities obtained from a stated variational problem, boundary-value problem, or theorem without using the empirical fine-structure constant, CODATA, m_e , e , R_∞ , or $\hbar$ as numerical inputs. Under this convention the local logarithmic coefficient $A_K \rightarrow 1/(4\pi)$ is derived, but the pressure-cell coefficients introduced below are not yet first-principles derived. They are structured closure hypotheses.

Table 1: Status of the pressure-cell ingredients in this paper.

Item	Expression	Current status	Requirement for “derived” label
Local logarithmic coefficient	$A_K \rightarrow 1/(4\pi)$	Theorem	Already derived from the local Biot–Savart
Zeroth pressure prefactor	$16\pi/3$	Closure hypothesis	Derive from a specified spherical pressure-
Finite-shell correction	$11/(48\mathcal{L}_K^2)$	Closure hypothesis	Derive from a controlled NLS/GP core-pr
Cell radius closure	$R_{\text{cell}} = 2L_K$	Closure hypothesis	Derive from stationarity or include χ_R -ser

Thus this paper supplies a coherent analytical closure framework. It does not by itself prove that the electromagnetic fine-structure constant is derived from first principles. That stronger label requires deriving the entries marked as closure hypotheses in Table 1.

3 Finite-core filament energy

Consider a smooth closed filament centreline $X(s) \subset \mathbb{R}^3$, parametrized by arclength $s \in [0, L_K]$, with unit tangent $T(s) = \partial_s X(s)$. Let a be a small core cutoff satisfying

$$a \ll \ell_{\text{curv}}, \quad a \ll d_{\text{min}},$$

where ℓ_{curv} is the local curvature scale and d_{min} is the minimum nonlocal self-distance of the centreline.

The dimensionless cutoff Biot–Savart energy used in the numerical implementation is

$$E_{\text{BS}}^{\text{norm}}(a) = \frac{1}{8\pi} \int_0^{L_K} \int_0^{L_K} \frac{T(s) \cdot T(s')}{\|X(s) - X(s')\|} \Theta(\|X(s) - X(s')\| - a) \, ds \, ds', \quad (1)$$

where Θ is a cutoff indicator. Its physical normalization is

$$E_{\text{BS}}(a) = \rho_f \Gamma_0^2 E_{\text{BS}}^{\text{norm}}(a). \quad (2)$$

For a slender filament, the fitted local form is

$$E_{\text{BS}}(a) = \rho_f \Gamma_0^2 L_K \left[A_K \log\left(\frac{L_K}{a}\right) + a_K \right] + o(1) \quad (a/L_K \rightarrow 0). \quad (3)$$

4 Local logarithmic coefficient

Theorem 1 (Universal local Biot–Savart logarithm). *For any C^2 smooth slender closed filament, the leading local logarithmic coefficient in Eq. (3) is*

$$A_K = \frac{1}{4\pi}$$

provided the energy is normalized as in Eq. (1).

Proof. Fix s and write $u = s' - s$. In the near-diagonal region $a < |u| < \ell$, where $a \ll \ell \ll \ell_{\text{curv}}$, smoothness gives

$$X(s+u) - X(s) = uT(s) + O(u^2), \quad T(s) \cdot T(s+u) = 1 + O(u^2).$$

Therefore

$$\frac{T(s) \cdot T(s+u)}{\|X(s+u) - X(s)\|} = \frac{1}{|u|} + O(1).$$

The local singular part of Eq. (1) per unit arclength is then

$$\frac{1}{8\pi} \int_{a < |u| < \ell} \frac{du}{|u|} = \frac{1}{8\pi} \left[2 \log\left(\frac{\ell}{a}\right) \right] = \frac{1}{4\pi} \log\left(\frac{\ell}{a}\right).$$

Nonlocal contributions and curvature-dependent finite terms alter a_K , not the coefficient of $\log(1/a)$. Hence $A_K = 1/(4\pi)$. $\square$

Remark 1. *This theorem is local. It does not assert that every finite-resolution numerical plateau is exactly $1/(4\pi)$; it asserts the slender-core asymptotic coefficient. Knot geometry enters the finite part a_K , the ropelength $\mathcal{L}_K$, and nonlocal corrections.*

5 Pressure balance and core closure

The normal pressure exerted by the exterior irrotational self-energy on a tube of radius a is the negative derivative with respect to volume. Since the tube area is $2\pi a L_K$,

$$P_{\text{irr}}(a) = -\frac{1}{2\pi a L_K} \frac{\partial E_{\text{BS}}}{\partial a}. \quad (4)$$

Using Eq. (3),

$$P_{\text{irr}}(a) = \frac{\rho_f \Gamma_0^2 A_K}{2\pi a^2}. \quad (5)$$

The core kinetic pressure scale is

$$P_{\text{core}} = \frac{1}{2} \rho_f \|\mathbf{v}_\odot\|^2. \quad (6)$$

The no-contact pressure balance $P_{\text{irr}} = P_{\text{core}}$ gives

$$\frac{\rho_f \Gamma_0^2 A_K}{2\pi a^2} = \frac{1}{2} \rho_f \|\mathbf{v}_\odot\|^2, \quad (7)$$

hence

$$a = \frac{\Gamma_0}{\|\mathbf{v}_\odot\|} \sqrt{\frac{A_K}{\pi}}. \quad (8)$$

With the circulation relation

$$\Gamma_0 = 2\pi r_c \|\mathbf{v}_\odot\|, \quad (9)$$

this becomes

$$\boxed{\frac{a_{\text{nc}}}{r_c} = \sqrt{4\pi A_K}}. \quad (10)$$

Therefore, in the theorem limit $A_K = 1/(4\pi)$,

$$\boxed{a_{\text{nc}} = r_c}. \quad (11)$$

Dimensional consistency is immediate:

$$[P_{\text{irr}}] = \frac{[\rho_f][\Gamma_0]^2}{[a]^2} = \frac{\text{kg m}^{-3} \text{ m}^4 \text{ s}^{-2}}{\text{m}^2} = \text{kg m}^{-1} \text{ s}^{-2} = \text{Pa}.$$

6 Spherical pressure-cell functional

The local calculation fixes the tube radius. The next question is whether the exterior spherical cell can generate the finite correction required for the fine-structure scale.

Let a trefoil tube of total length L_K and diameter $D = 2a$ be embedded in a spherical coarse-graining cell of radius

$$R_{\text{cell}} = 2L_K. \quad (12)$$

Then

$$\eta_K = \frac{a}{R_{\text{cell}}} = \frac{a}{2L_K} = \frac{1}{4(L_K/D)} = \frac{1}{4\mathcal{L}_K}. \quad (13)$$

6.1 Geometric pressure-transfer factor

A small outward radial displacement of the spherical control surface by a gives the surface-area Jacobian

$$J_A(\eta_K) = \frac{4\pi(R_{\text{cell}} + a)^2}{4\pi R_{\text{cell}}^2} = (1 + \eta_K)^2 = 1 + 2\eta_K + \eta_K^2. \quad (14)$$

The isotropic radial volume-response mode contributes one additional linear factor,

$$J_V^{(r)}(\eta_K) = \eta_K. \quad (15)$$

The total spherical pressure-transfer multiplier is therefore

$$\boxed{\Xi_{\text{sph}} = J_A + J_V^{(r)} = 1 + 3\eta_K + \eta_K^2 = 1 + \frac{3}{4\mathcal{L}_K} + \frac{1}{16\mathcal{L}_K^2}.} \quad (16)$$

6.2 Explicit variational form

To make the above statement variational rather than merely kinematic, introduce an isotropic cell-response coordinate q . Let $K_{\text{sph}} > 0$ be a spherical compression modulus and define

$$\mathcal{G}_{\text{sph}}(q; \eta_K) = \frac{K_{\text{sph}}}{2}q^2 - K_{\text{sph}} \left[J_A(\eta_K) + J_V^{(r)}(\eta_K) \right] q. \quad (17)$$

Stationarity gives

$$\frac{\partial \mathcal{G}_{\text{sph}}}{\partial q} = K_{\text{sph}} \left[q - \left(J_A + J_V^{(r)} \right) \right] = 0, \quad (18)$$

hence

$$q_{\star} = \Xi_{\text{sph}}. \quad (19)$$

Since

$$\frac{\partial^2 \mathcal{G}_{\text{sph}}}{\partial q^2} = K_{\text{sph}} > 0, \quad (20)$$

the stationary point is a strict minimum.

Remark 2. Equation (17) is an explicit coarse-grained pressure-cell functional. It does not yet derive $R_{\text{cell}} = 2L_K$ from primitive dynamics; Eq. (12) is a closure choice. A fully fundamental derivation must replace Eq. (12) by a stationarity condition on R_{cell} .

The derivation-target lemmas for the pressure prefactor, the finite-shell coefficient, and the radius closure are collected in Appendix C. This appendix is part of the derived-label gate: it identifies the exact mode-count and shell-variation results still required to upgrade the pressure-cell scale from structured closure to first-principles derivation.

7 NLS finite-shell pressure scale

The spherical response factor in Sec. 6 fixes the finite-core transfer geometry, but the companion finite-cell eigenvalue problem also requires a pressure equilibrium scale and a stiffness. The non-linear Schrödinger/Gross–Pitaevskii vortex-core asymptotics provide a natural source for both quantities. For a large vortex ring with core radius r_c , the standard asymptotic form is

$$E_{\text{ring}}(R) = \frac{1}{2}\rho\Gamma^2R\left[\ln\left(\frac{8R}{r_c}\right) - A\right], \quad (21)$$

and the translational velocity has the form

$$V_{\text{ring}}(R) = \frac{\Gamma}{4\pi R}\left[\ln\left(\frac{8R}{r_c}\right) - B\right]. \quad (22)$$

Hamiltonian consistency $V_{\text{ring}} = dE_{\text{ring}}/dP$, with $P(R) = \rho\Gamma\pi R^2$, gives

$$B = A - 1. \quad (23)$$

The Golden NLS branch uses

$$A = \varphi, \quad B = \varphi^{-1}, \quad \varphi = \frac{1 + \sqrt{5}}{2}. \quad (24)$$

The zeroth-order spherical pressure scale associated with an ideal trefoil cell is

$$E_p^{(0)} = \frac{16\pi}{3}\mathcal{L}_K. \quad (25)$$

Finite tube thickness contributes a second-order shell correction in the small parameter

$$\eta_K = \frac{1}{4\mathcal{L}_K}. \quad (26)$$

The NLS finite-shell pressure scale used in the companion BEM calculation is

$$E_p^{\text{NLS}} = \frac{16\pi}{3}\mathcal{L}_K\left[1 - \frac{11}{48\mathcal{L}_K^2}\right] = \frac{16\pi}{3}\mathcal{L}_K\left[1 - \frac{11}{3}\eta_K^2\right]. \quad (27)$$

The corresponding shell stiffness is

$$\kappa_{\text{NLS}}^{\text{shell}} = 8\pi\mathcal{L}_K^2 = \frac{\pi}{2\eta_K^2}. \quad (28)$$

The coefficient 11/48 is used here as a finite-shell closure coefficient, not as a fit to α . At the present stage it should be read as a structured NLS-shell ansatz. A first-principles derivation would require a controlled GP/NLS core-profile or shell expansion that produces this coefficient without reference to the empirical fine-structure constant. The stiffness $\kappa_{\text{NLS}}^{\text{shell}}$ is the corresponding thin-shell stiffness; it is large because $\eta_K \ll 1$.

The finite-cell pressure enthalpy used by the companion BEM eigenvalue calculation is

$$\mathcal{A}_{\text{pressure}}(E) = \kappa_{\text{NLS}}^{\text{shell}}\left[\frac{1}{3}\left(\frac{E}{E_p^{\text{NLS}}}\right)^3 - \ln\left(\frac{E}{E_p^{\text{NLS}}}\right)\right]. \quad (29)$$

It satisfies

$$\frac{d\mathcal{A}_{\text{pressure}}}{d\log E} = \kappa_{\text{NLS}}^{\text{shell}}\left[\left(\frac{E}{E_p^{\text{NLS}}}\right)^3 - 1\right], \quad (30)$$

so the pressure part alone has a strict stationary point at $E = E_p^{\text{NLS}}$. The BEM part in the companion paper supplies the finite-cell compliance correction:

$$\mathcal{A}_{\text{total}}(E) = \mathcal{A}_{\text{BEM}}(E) + \mathcal{A}_{\text{pressure}}(E), \quad \frac{d\mathcal{A}_{\text{total}}}{d \log E} = 0. \quad (31)$$

Thus the present paper supplies the analytical NLS-shell closure, while the companion calculation determines the stationary eigenvalue E_* .

For the ideal trefoil value $\mathcal{L}_K = 16.371637$, Eq. (27) gives

$$E_p^{(0)} = 274.309410808, \quad (32)$$

$$E_p^{\text{NLS}} = 274.074875657, \quad (33)$$

$$\kappa_{\text{NLS}}^{\text{shell}} = 6736.341149. \quad (34)$$

These are dimensionless geometric/NLS closure quantities; no electron mass, Planck constant, charge, Rydberg constant, or CODATA value is used to evaluate them. Their use as first-principles inputs remains conditional on deriving the prefactor $16\pi/3$, the shell coefficient $11/48$, and the cell-radius closure.

8 Aspect renormalization and fine-structure scale

The pressure-cell closure assumes that the zeroth-order aspect scale $\mathcal{E}_0$ receives a small finite-core correction proportional to the local core coefficient and to the spherical transfer response. In the theorem limit $A_K = 1/(4\pi)$,

$$\pi^2 A_K \rightarrow \frac{\pi}{4}. \quad (35)$$

The effective aspect scale is therefore

$$\mathcal{E}_{\text{eff}} = \mathcal{E}_0 \left[1 - \frac{\pi}{4\mathcal{E}_0^2} \Xi_{\text{sph}} \right]. \quad (36)$$

The closure identification is

$$\alpha_{\text{cell}} = \frac{2}{\mathcal{E}_{\text{eff}}}. \quad (37)$$

Combining Eqs. (16), (36), and (37) gives

$$\alpha_{\text{cell}} = \frac{2}{\mathcal{E}_0 \left[1 - \frac{\pi}{4\mathcal{E}_0^2} \left(1 + \frac{3}{4\mathcal{L}_K} + \frac{1}{16\mathcal{L}_K^2} \right) \right]}. \quad (38)$$

Remark 3. Equation (38) is a theorem inside the specified closure model. It is not yet an unconditional derivation of the QED coupling because the far-field interaction law has not been derived. The necessary additional theorem is stated in Sec. 10.

9 Numerical evaluation

Use the ideal-trefoil ropelength value

$$\mathcal{L}_K = 16.371637, \quad (39)$$

taken from the ideal-knot Fourier data source. Then

$$\eta_K = \frac{1}{4\mathcal{L}_K} = 0.01527031170, \quad (40)$$

$$\Xi_{\text{sph}} = 1 + 3\eta_K + \eta_K^2 = 1.046044118, \quad (41)$$

$$E_p^{(0)} = \frac{16\pi}{3}\mathcal{L}_K = 274.309410808, \quad (42)$$

$$E_p^{\text{NLS}} = \frac{16\pi}{3}\mathcal{L}_K \left(1 - \frac{11}{48\mathcal{L}_K^2}\right) = 274.074875657, \quad (43)$$

$$\kappa_{\text{NLS}}^{\text{shell}} = 8\pi\mathcal{L}_K^2 = 6736.341149. \quad (44)$$

The companion BEM/NLS calculation uses E_p^{NLS} and $\kappa_{\text{NLS}}^{\text{shell}}$ in Eq. (31) and finds the stationary aspect eigenvalue $E_\star$. Once that eigenvalue is supplied, the finite-core effective aspect is evaluated as

$$E_{\text{eff}} = E_\star \left[1 - \frac{\pi}{4E_\star^2} \left(1 + \frac{3}{4\mathcal{L}_K} + \frac{1}{16\mathcal{L}_K^2}\right)\right], \quad (45)$$

and the dimensionless fine-structure-scale estimate is

$$\alpha_{\text{cell}} = \frac{2}{E_{\text{eff}}}. \quad (46)$$

Table 2: Analytical pressure-cell quantities supplied to the companion BEM/NLS finite-cell eigenvalue calculation.

Quantity	Expression	Value
Ideal trefoil ropelength	$\mathcal{L}_K$	16.371637
Shell parameter	$\eta_K = 1/(4\mathcal{L}_K)$	0.01527031170
Spherical transfer	Ξ_{sph}	1.046044118
Zeroth pressure scale	$E_p^{(0)} = (16\pi/3)\mathcal{L}_K$	274.309410808
NLS pressure scale	E_p^{NLS}	274.074875657
NLS shell stiffness	$\kappa_{\text{NLS}}^{\text{shell}} = 8\pi\mathcal{L}_K^2$	6736.341149

10 What remains for a fundamental derivation

The preceding sections provide a conditional mathematical closure. To upgrade it to a fundamental derivation, the following results must be obtained without using α , R_∞ , e , or CODATA as input.

10.1 Open Theorem I: topological aspect eigenvalue

The number

$$\mathcal{E}_0 = 274.074996 \quad (47)$$

must be obtained as an eigenvalue of a well-posed geometric or hydrodynamic variational problem:

$$\delta\mathcal{F}[X, \mathbf{u}, p, R_{\text{cell}}, a] = 0, \quad \nabla \cdot \mathbf{u} = 0, \quad (48)$$

with trefoil topology and circulation constraints. Without this theorem, $\mathcal{E}_0$ remains an external aspect input.

10.2 Open Theorem II: spherical radius from stationarity

The closure radius $R_{\text{cell}} = 2L_K$ should be replaced by

$$\frac{\partial \mathcal{F}}{\partial R_{\text{cell}}} = 0. \quad (49)$$

If this stationarity condition yields $R_{\text{cell}} = 2L_K$, then $\eta_K = 1/(4\mathcal{L}_K)$ becomes dynamical rather than chosen.

10.3 Open Theorem III: far-field interaction law

A fine-structure constant is not only a number. In conventional electrodynamics it is the coefficient of a long-range coupling. The required far-field theorem is

$$V_{\text{eff}}(r) = \frac{\alpha_{\text{cell}} \hbar c}{r} + O(r^{-2}) \quad (r \rightarrow \infty). \quad (50)$$

Equivalently, the model must yield

$$\alpha_{\text{cell}} = \frac{g_{\text{eff}}^2}{4\pi \hbar c}. \quad (51)$$

Only after Eq. (50) is derived from the same variational structure can Eq. (38) be called a fundamental derivation of the electromagnetic coupling.

11 Falsification criteria

The closure chain is falsified, or at least not fundamental, if any of the following fail:

- (F1) The slender-core coefficient does not converge to $A_K = 1/(4\pi)$.
- (F2) The ideal-trefoil ropelength used for $\mathcal{L}_K$ is not the relevant minimizing geometry.
- (F3) A pressure-cell variational problem does not yield $R_{\text{cell}} = 2L_K$ or an equivalent $\eta_K = 1/(4\mathcal{L}_K)$.
- (F4) The aspect-renormalization law Eq. (36) cannot be derived from a more primitive pressure or action principle.
- (F5) The far-field interaction is not of the $1/r$ form in Eq. (50).

12 Conclusion

The pressure-coefficient gates are stated explicitly in Appendix C; until those gates are closed, the NLS pressure scale should be called closure-derived rather than first-principles derived.

The mathematical route from calibration to pressure-cell closure is now separated into three levels. First, the local Biot–Savart singularity gives the slender-core coefficient $A_K \rightarrow 1/(4\pi)$, and the pressure balance then fixes $a_{\text{nc}} = r_c$. Second, the spherical control-volume response gives

$$\Xi_{\text{sph}} = 1 + \frac{3}{4\mathcal{L}_K} + \frac{1}{16\mathcal{L}_K^2}.$$

Third, the NLS finite-shell closure supplies the pressure scale and stiffness

$$E_p^{\text{NLS}} = \frac{16\pi}{3} \mathcal{L}_K \left(1 - \frac{11}{48\mathcal{L}_K^2} \right), \quad \kappa_{\text{NLS}}^{\text{shell}} = 8\pi \mathcal{L}_K^2.$$

For the ideal trefoil these evaluate to

$$E_p^{\text{NLS}} = 274.074875657, \quad \kappa_{\text{NLS}}^{\text{shell}} = 6736.341149.$$

These quantities are computed from the ropelength geometry and the stated NLS-shell closure hypotheses, not from α , m_e , $\hbar$, e , or R_∞ . They should therefore be labelled as closure-derived rather than first-principles derived until the coefficients in Table 1 are obtained from an independent variational calculation. The stationary aspect eigenvalue $E_\star$ is obtained in the companion BEM/NLS finite-cell calculation. The final identification with the electromagnetic fine-structure constant further requires the far-field interaction theorem

$$V(r) = \frac{\alpha_{\text{cell}} \hbar c}{r} + O(r^{-2}).$$

A Dimensional checks

The three key energy terms have dimensions

$$[\rho_f \Gamma_0^2 L_K] = \text{kg m}^{-3} \text{m}^4 \text{s}^{-2} \text{m} = \text{J}, \quad (52)$$

$$\left[\frac{1}{2} \pi \rho_f \|\mathbf{v}_\odot\|^2 a^2 L_K \right] = \text{kg m}^{-3} \text{m}^2 \text{s}^{-2} \text{m}^2 \text{m} = \text{J}, \quad (53)$$

$$[P_{\text{irr}}] = \text{Pa}. \quad (54)$$

The quantities A_K , $\mathcal{L}_K$, η_K , Ξ_{sph} , $\mathcal{E}_0$, $\mathcal{E}_{\text{eff}}$, and α_{cell} are dimensionless.

B Small-parameter expansion

Let

$$\epsilon = \frac{1}{\mathcal{E}_0}.$$

Then Eq. (36) is

$$\mathcal{E}_{\text{eff}} = \mathcal{E}_0 \left[1 - \frac{\pi}{4} \Xi_{\text{sph}} \epsilon^2 \right].$$

Therefore

$$\alpha_{\text{cell}} = \frac{2}{\mathcal{E}_0} \left[1 + \frac{\pi}{4} \Xi_{\text{sph}} \epsilon^2 + O(\epsilon^4) \right].$$

Since $\epsilon \simeq 3.65 \times 10^{-3}$, the correction is naturally of order 10^{-5} , matching the observed difference between the zeroth-order $2/\mathcal{E}_0$ estimate and the refined closure.

C Derivation-target lemmas for pressure-cell coefficients

This appendix records the exact gates that must be closed before the pressure-cell scale can be labelled first-principles derived. The statements below are intentionally split into algebraic implications, which are proved, and physical inputs, which remain derivation targets.

C.1 Lemma A1: spherical pressure-volume factor

Let $\mathcal{L}_K = L_K/D$ be the ideal-trefoil ropelength. Suppose the zeroth-order pressure-cell action decomposes into N_p equivalent spherical pressure sectors, each contributing the normalized spherical volume factor $4\pi/3$ per unit ropelength. Then

$$E_p^{(0)} = N_p \frac{4\pi}{3} \mathcal{L}_K. \quad (55)$$

Consequently,

$$E_p^{(0)} = \frac{16\pi}{3} \mathcal{L}_K \iff N_p = 4. \quad (56)$$

Proof. Equating Eq. (55) to the target coefficient gives

$$N_p \frac{4\pi}{3} = \frac{16\pi}{3},$$

hence $N_p = 4$. $\square$

Derived-label gate. The coefficient $16\pi/3$ is derived only after $N_p = 4$ is derived from a pressure-cell variational principle or an independent boundary-mode count. In the present paper $N_p = 4$ is therefore a structured closure target.

C.2 Lemma A2: NLS finite-shell second variation

Let

$$\eta_K = \frac{a}{R_{\text{cell}}} = \frac{1}{2\chi_R \mathcal{L}_K}, \quad R_{\text{cell}} = \chi_R L_K, \quad (57)$$

where $D = 2a$. Assume a second-order shell response

$$E_p^{\text{shell}} = E_p^{(0)} [1 - \sigma \eta_K^2 + O(\eta_K^3)]. \quad (58)$$

Then

$$E_p^{\text{shell}} = E_p^{(0)} \left[1 - \frac{\sigma}{4\chi_R^2 \mathcal{L}_K^2} + O(\mathcal{L}_K^{-3}) \right]. \quad (59)$$

For $\chi_R = 2$, the coefficient $11/(48\mathcal{L}_K^2)$ is obtained iff

$$\sigma = \frac{11}{3}. \quad (60)$$

Proof. Substitution of Eq. (57) into Eq. (58) gives

$$\sigma \eta_K^2 = \frac{\sigma}{4\chi_R^2 \mathcal{L}_K^2}.$$

At $\chi_R = 2$, this is $\sigma/(16\mathcal{L}_K^2)$. Matching

$$\frac{\sigma}{16} = \frac{11}{48}$$

gives $\sigma = 11/3$. $\square$

A concrete NLS-shell target is the decomposition

$$\sigma = 3 + \frac{2}{3} = \frac{11}{3}. \quad (61)$$

The first contribution is assigned to the second variation of the spherical volume/pressure response. The second contribution is assigned to isotropic transverse shell averaging, e.g.

$$\langle \sin^2 \theta \rangle_{S^2} = \frac{2}{3}.$$

A future GP/NLS core-profile calculation must produce this decomposition without reference to the empirical fine-structure constant.

C.3 Minimal radius-closure target

The current pressure-cell construction uses

$$R_{\text{cell}} = 2L_K.$$

Introduce $\chi_R = R_{\text{cell}}/L_K$ and the minimal balancing action

$$A_\chi(\chi_R) = \chi_R + \frac{\lambda_\chi}{\chi_R}. \quad (62)$$

Then

$$\frac{dA_\chi}{d\chi_R} = 1 - \frac{\lambda_\chi}{\chi_R^2} = 0 \implies \chi_R = \sqrt{\lambda_\chi}. \quad (63)$$

Thus

$$\chi_R = 2 \iff \lambda_\chi = 4. \quad (64)$$

The open task is to derive $\lambda_\chi = 4$ from the inner/outer pressure-cell balance, rather than imposing $\chi_R = 2$.

C.4 Summary of Paper II derived-label gates

The pressure-cell route may be summarized as

$$\mathcal{L}_K \xrightarrow[\text{target: } N_p=4]{} \frac{16\pi}{3} \mathcal{L}_K \xrightarrow[\text{targets: } \sigma=11/3, \chi_R=2]{} E_p^{\text{NLS}}.$$

Thus Paper II supplies a structured closure framework. It becomes a first-principles pressure-scale derivation only after $N_p = 4$, $\sigma = 11/3$, and $\chi_R = 2$ are obtained from an independent pressure-shell variational calculation.

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